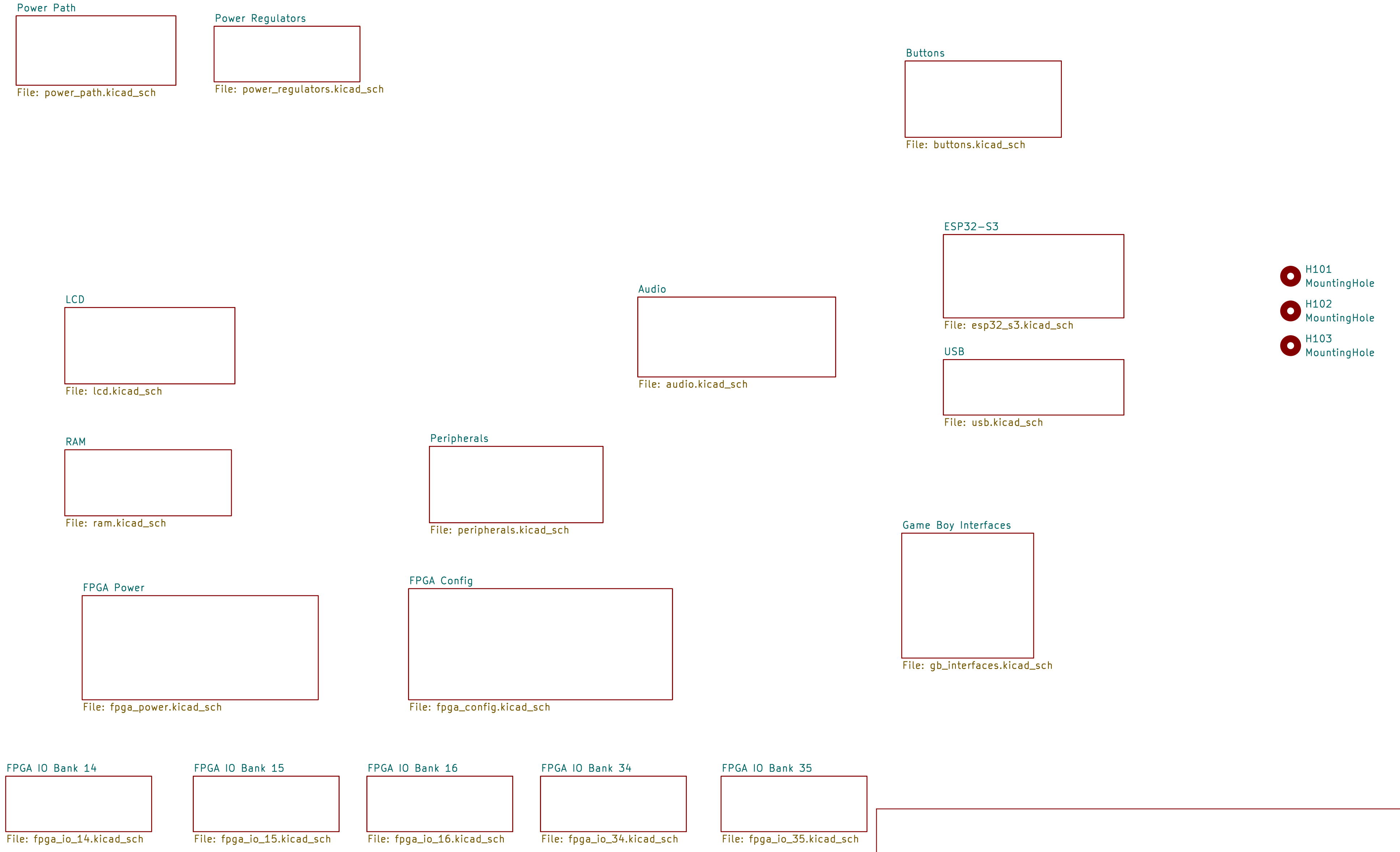
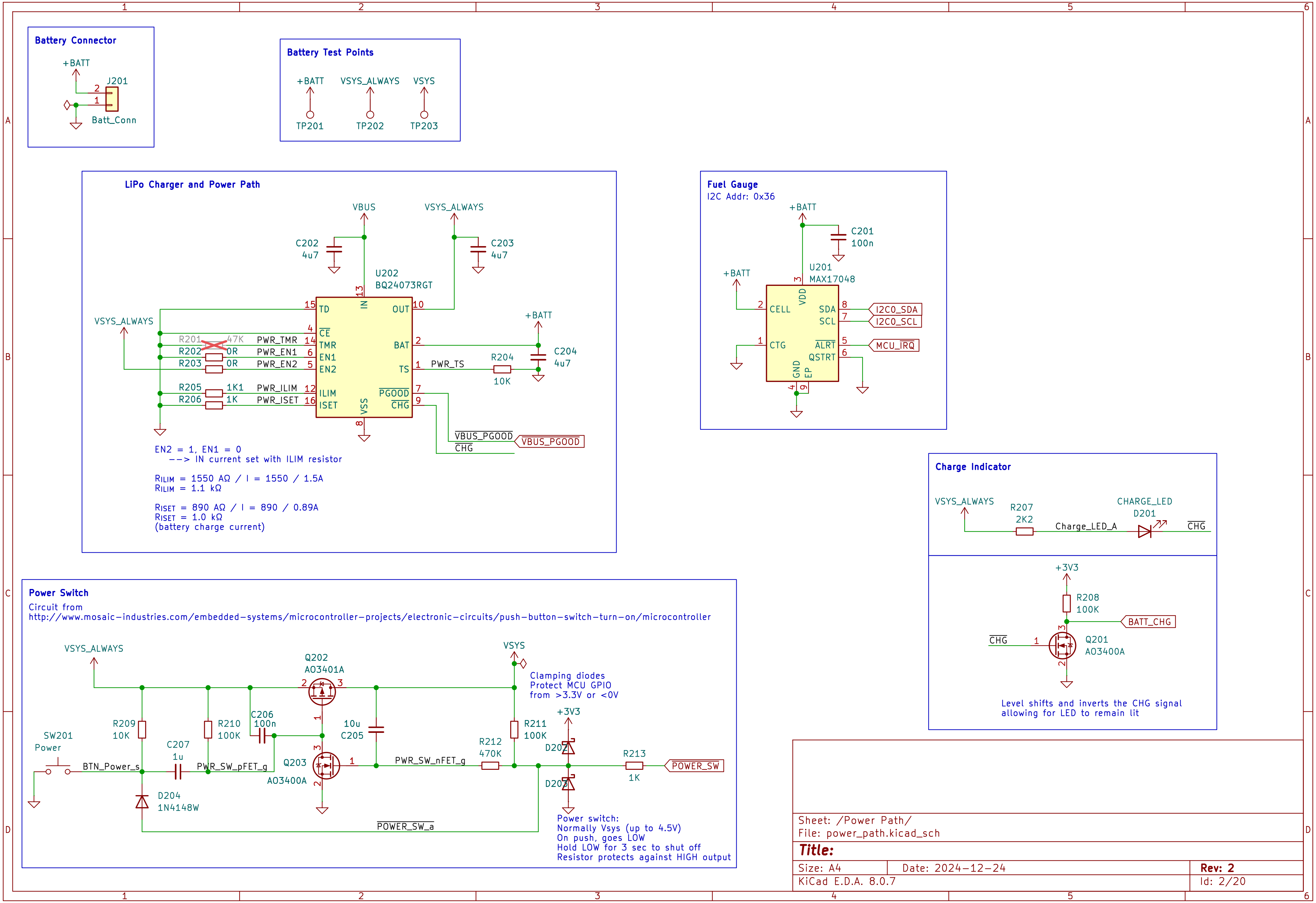
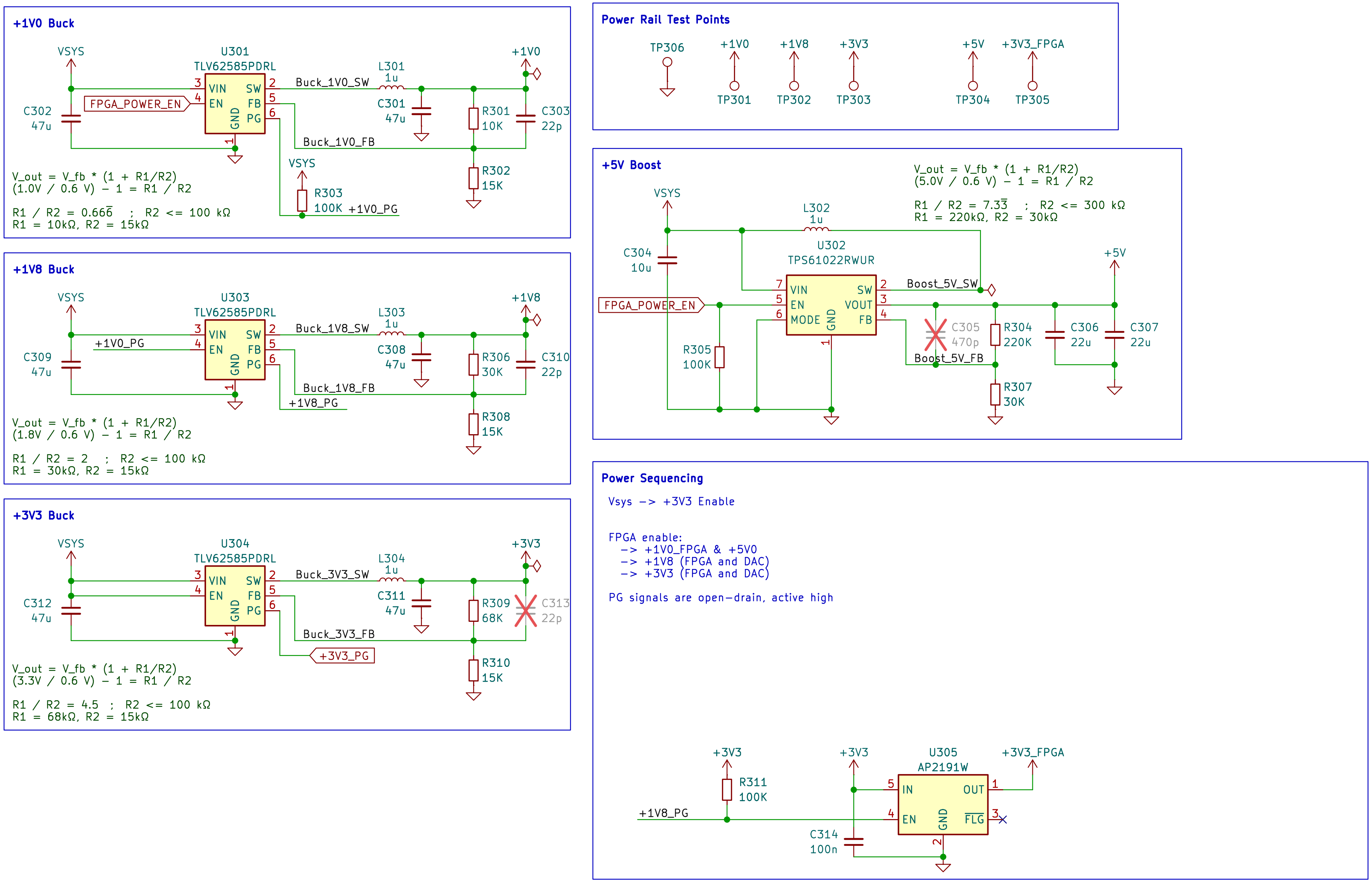


Sheet: /	
File: handheld.kicad_sch	
Title: Game Bub Handheld	
Size: A4 Date: 2024-12-24	**Rev: 2**
KiCad E.D.A. 8.0.7	Id: 1/20

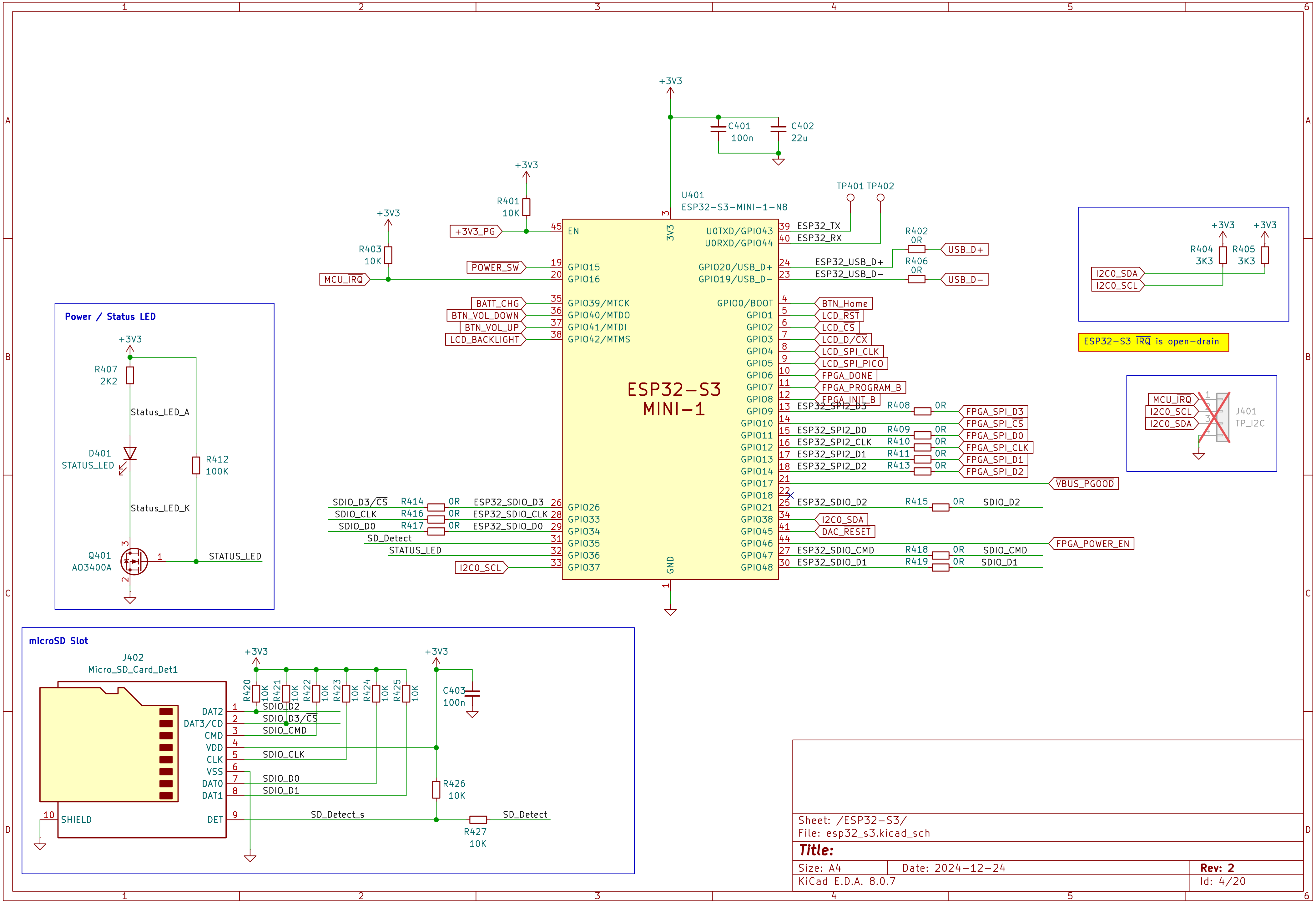

Battery Connector
+BATT
J201
Batt_Conn
Battery Test Points
+BATT
VSYS_ALWAYS
VSYS
TP201
TP202
TP203
LiPo Charger and Power Path
VBUS
VSYS_ALWAYS
C202 4u7
C203 4u7
U202
BQ24073RGT
TD
IN
OUT
CE
TMR
EN1
EN2
ILIM
ISET
VSS
BAT
TS
PGOOD
CHG
VSYS_ALWAYS
R201 47K
R202 0R
R203 0R
R205 1K1
R206 1K
PWR_TMR
PWR_EN1
PWR_EN2
PWR_ILIM
PWR_ISET
PWR_TS
R204 10K
+BATT
C204 4u7
VBUS_PGOOD
CHG
VBUS_PGOOD
EN2 = 1, EN1 = 0
--> IN current set with ILIM resistor
RILIM = 1550 AΩ / I = 1550 / 1.5A
RILIM = 1.1 kΩ
RISET = 890 AΩ / I = 890 / 0.89A
RISET = 1.0 kΩ
(battery charge current)
Fuel Gauge
I2C Addr: 0x36
+BATT
C201 100n
U201
MAX17048
+BATT
VDD
CELL
CTG
GND
EP
SDA
SCL
ALRT
QSTRT
I2C0_SDA
I2C0_SCL
MCU_IRQ
Charge Indicator
VSYS_ALWAYS
R207 2K2
Charge_LED_A
CHARGE_LED
D201
CHG
+3V3
R208 100K
BATT_CHG
CHG
Q201
AO3400A
Level shifts and inverts the CHG signal
allowing for LED to remain lit
Power Switch
Circuit from
http://www.mosaic-industries.com/embedded-systems/microcontroller-projects/electronic-circuits/push-button-switch-turn-on/microcontroller
VSYS_ALWAYS
Q202
AO3401A
VSYS
Clamping diodes
Protect MCU GPIO
from >3.3V or <0V
+3V3
R209 10K
R210 100K
C206 100n
10u C205
R211 100K
SW201 Power
BTN_Power_s
C207 1u
PWR_SW_pFET_g
Q203
AO3400A
PWR_SW_nFET_g
R212 470K
D202
D203
R213 1K
POWER_SW
D204 1N4148W
POWER_SW_a
Power switch:
Normally Vsys (up to 4.5V)
On push, goes LOW
Hold LOW for 3 sec to shut off
Resistor protects against HIGH output
Sheet: /Power Path/
File: power_path.kicad_sch
Title:
Size: A4
Date: 2024-12-24
Rev: 2
KiCad E.D.A. 8.0.7
Id: 2/20



Sheet: /Power Regulators/
File: power_regulators.kicad_sch

Title:

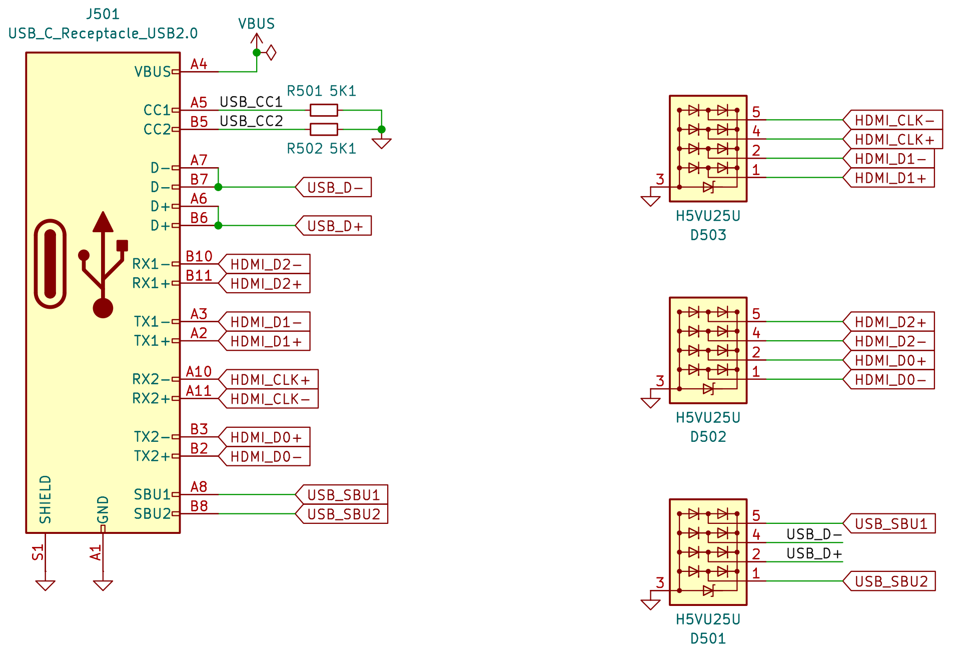
Size: A4	Date: 2024-12-24	**Rev: 2**
KiCad E.D.A. 8.0.7		Id: 3/20



Sheet: /ESP32-S3/
File: esp32_s3.kicad_sch

Title:

Size: A4	Date: 2024-12-24	**Rev: 2**
KiCad E.D.A. 8.0.7		Id: 4/20

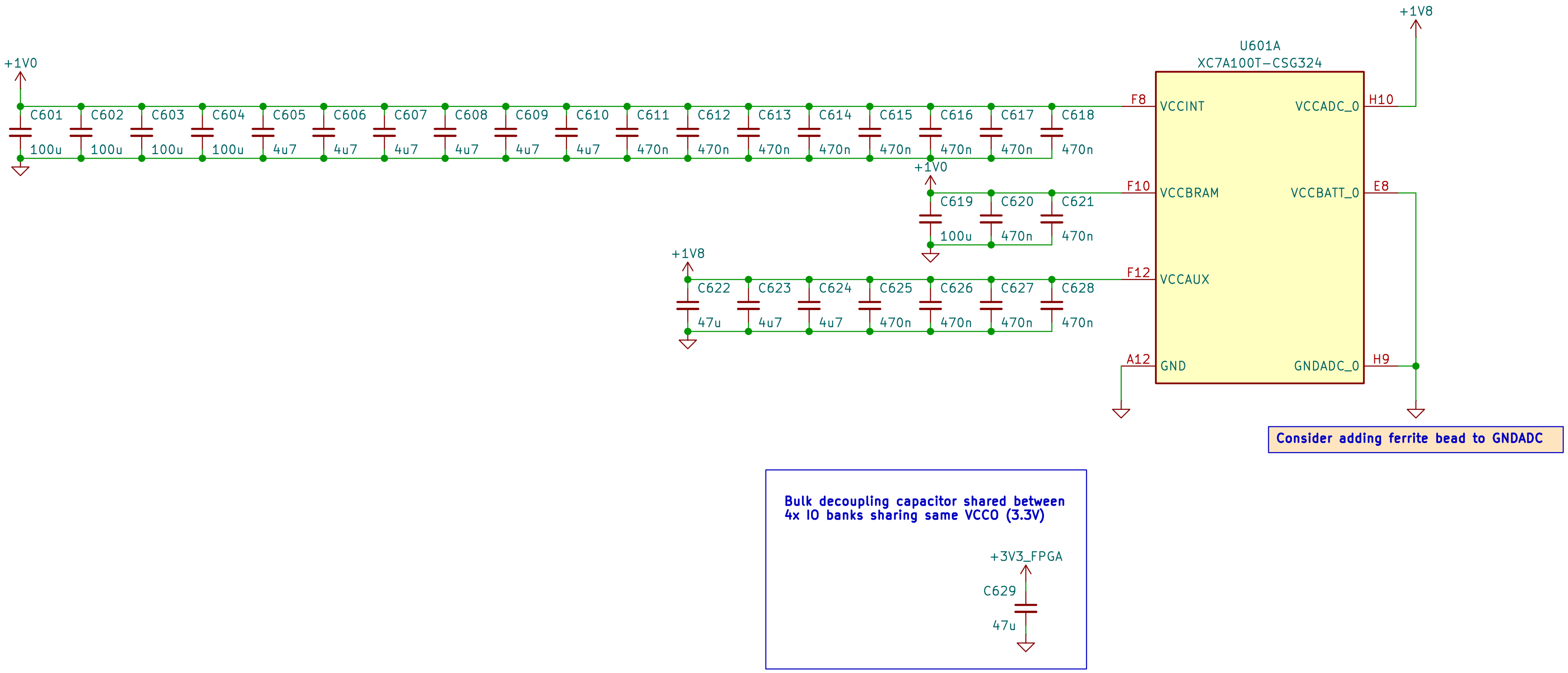


Sheet: /USB/
File: usb.kicad_sch

Title:

Size: A4 | Date: 2024-12-24 | **Rev: 2**

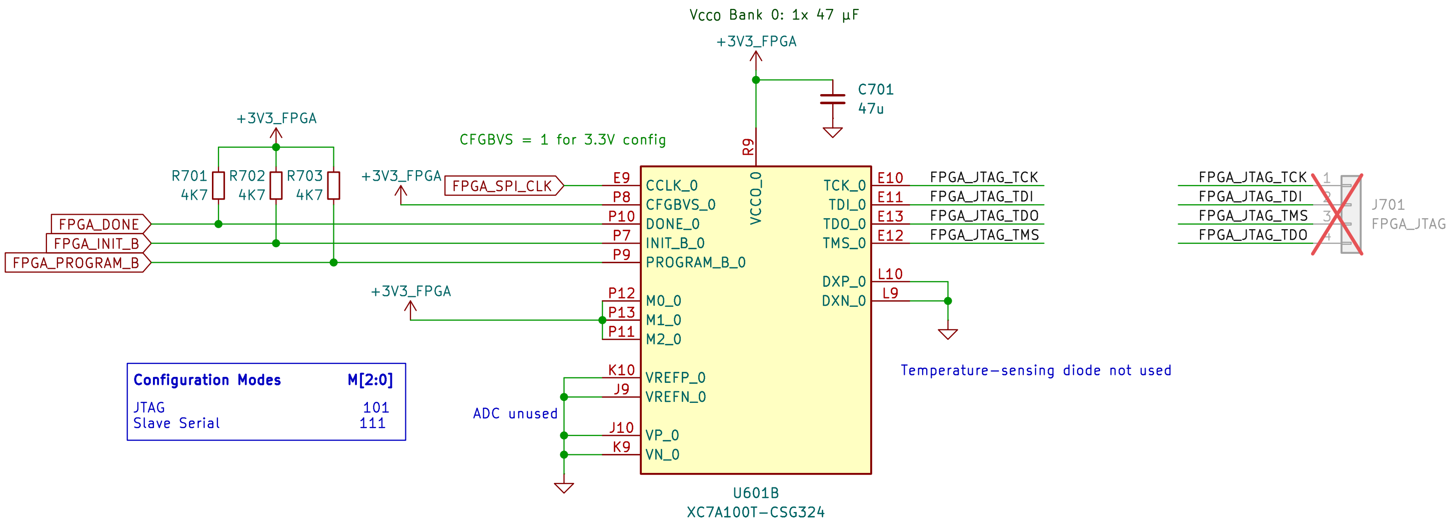
KiCad E.D.A. 8.0.7 | Id: 5/20

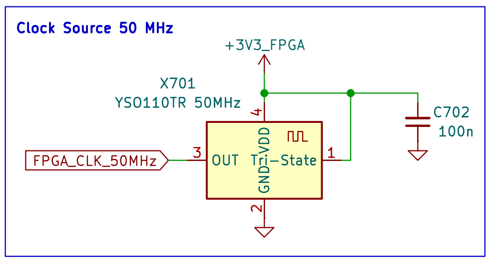


Sheet: /FPGA Power/
File: fpga_power.kicad_sch

Title:

Size: A4	Date: 2024-12-24	**Rev: 2**
KiCad E.D.A. 8.0.7		Id: 6/20

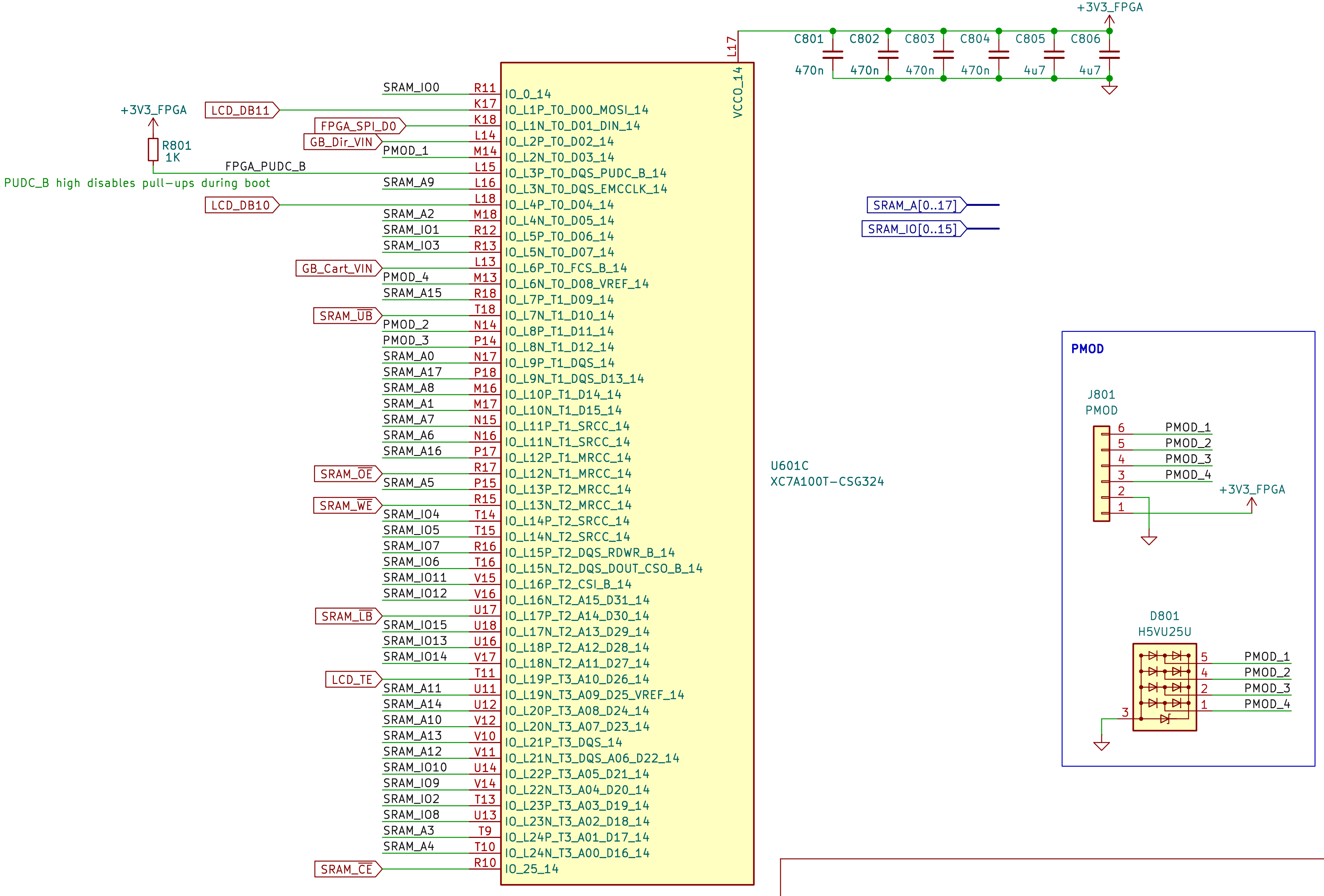




Sheet: /FPGA Config/
File: fpga_config.kicad_sch

Title:

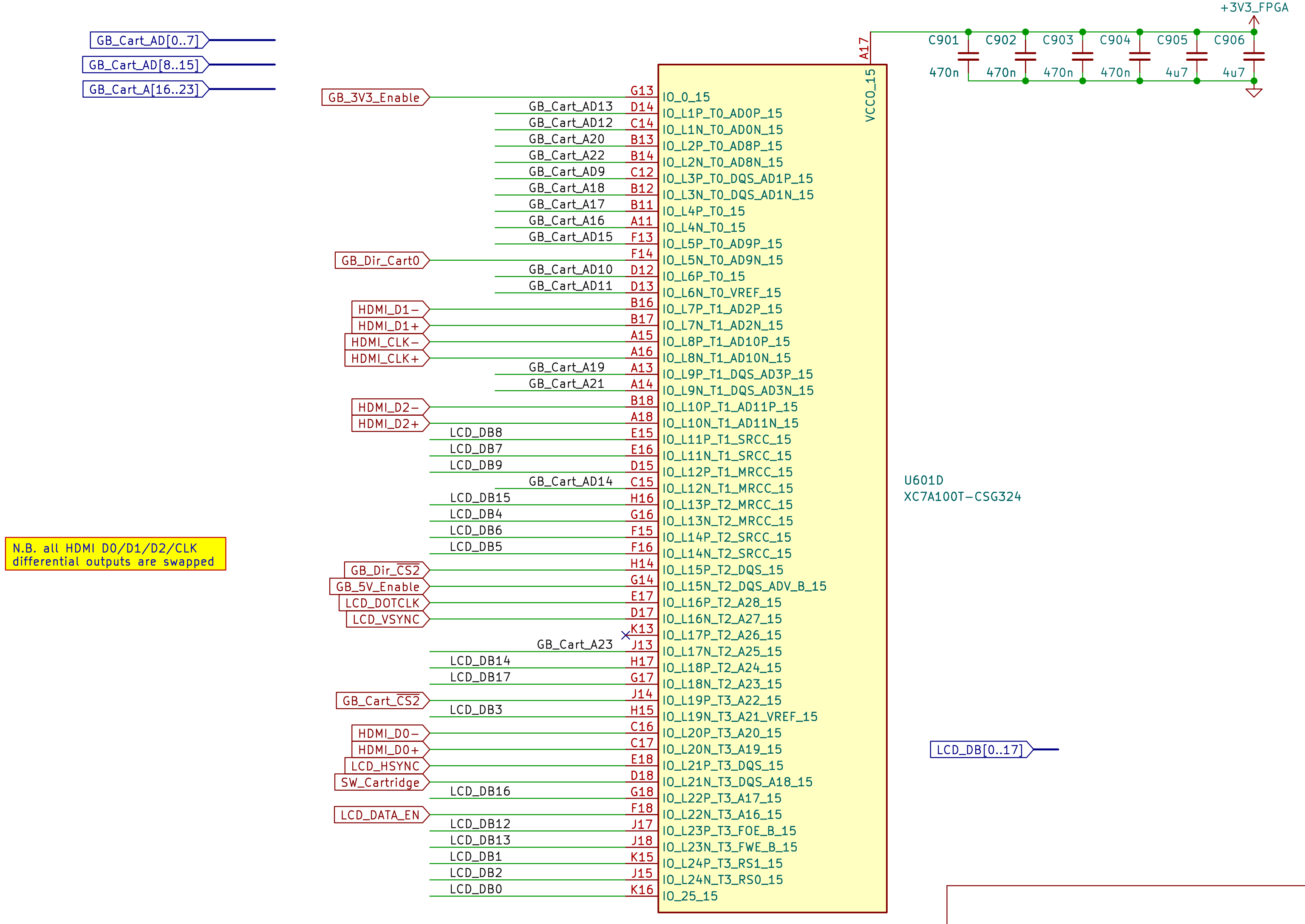
Size: A4	Date: 2024-12-24	**Rev: 2**
KiCad E.D.A. 8.0.7		Id: 6/20



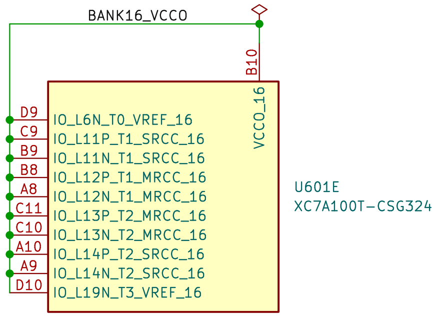
Sheet: /FPGA IO Bank 14/
File: fpga_io_14.kicad_sch

Title:

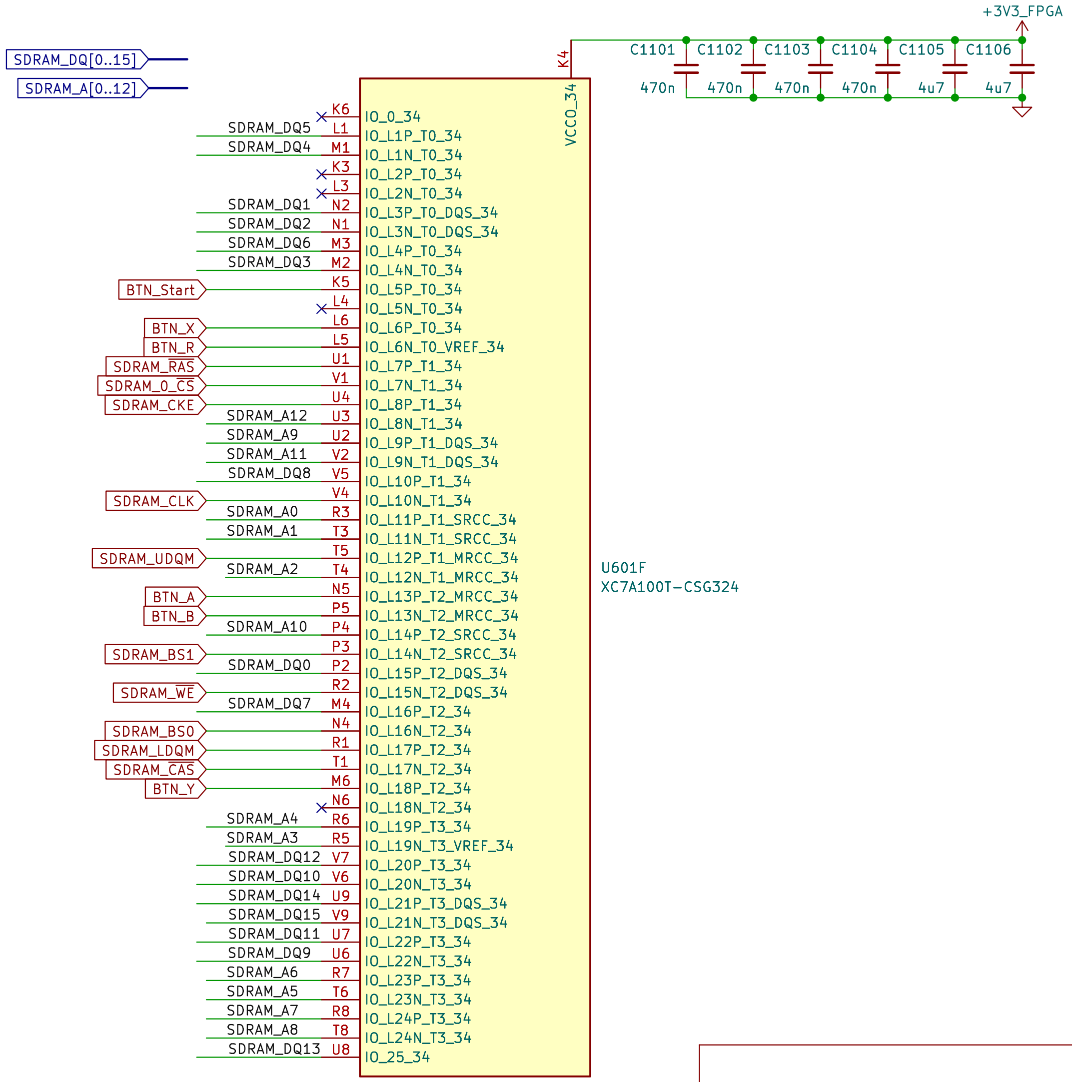
Size: A4	Date: 2024-12-24	**Rev: 2**
KiCad E.D.A. 8.0.7		Id: 7/20



Sheet: /FPGA IO Bank 15/		
File: fpga_io_15.kicad_sch		
Title:		
Size: A4	Date: 2024-12-24	**Rev: 2**
KiCad E.D.A. 8.0.7		Id: 8/20



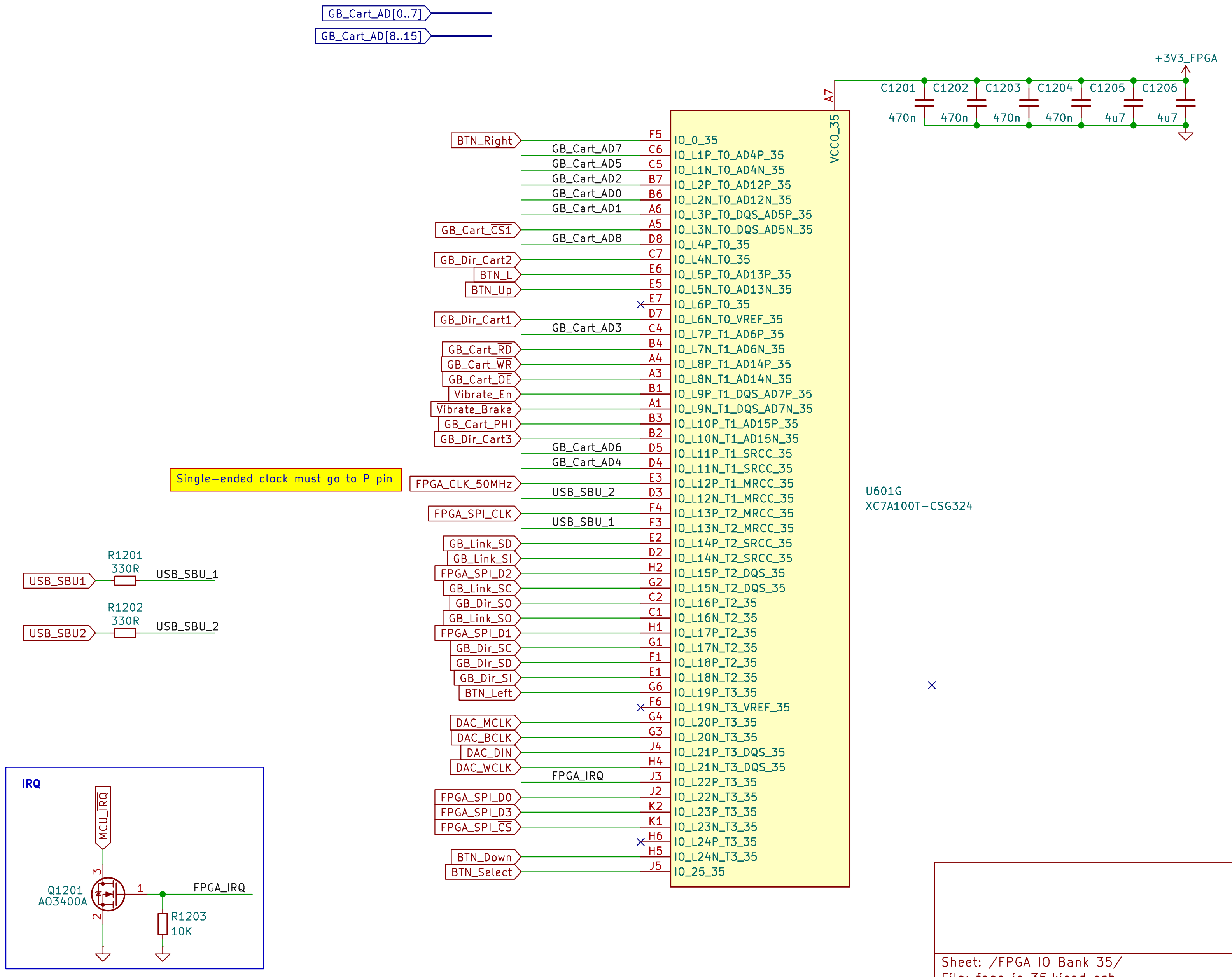
Sheet: /FPGA IO Bank 16/	
File: fpga_io_16.kicad_sch	
Title:	
Size: A4 Date: 2024-12-24	**Rev: 2**
KiCad E.D.A. 8.0.7	Id: 9/20



Sheet: /FPGA IO Bank 34/
File: fpga_io_34.kicad_sch

Title:

Size: A4	Date: 2024-12-24	**Rev: 2**
KiCad E.D.A. 8.0.7		Id: 10/20

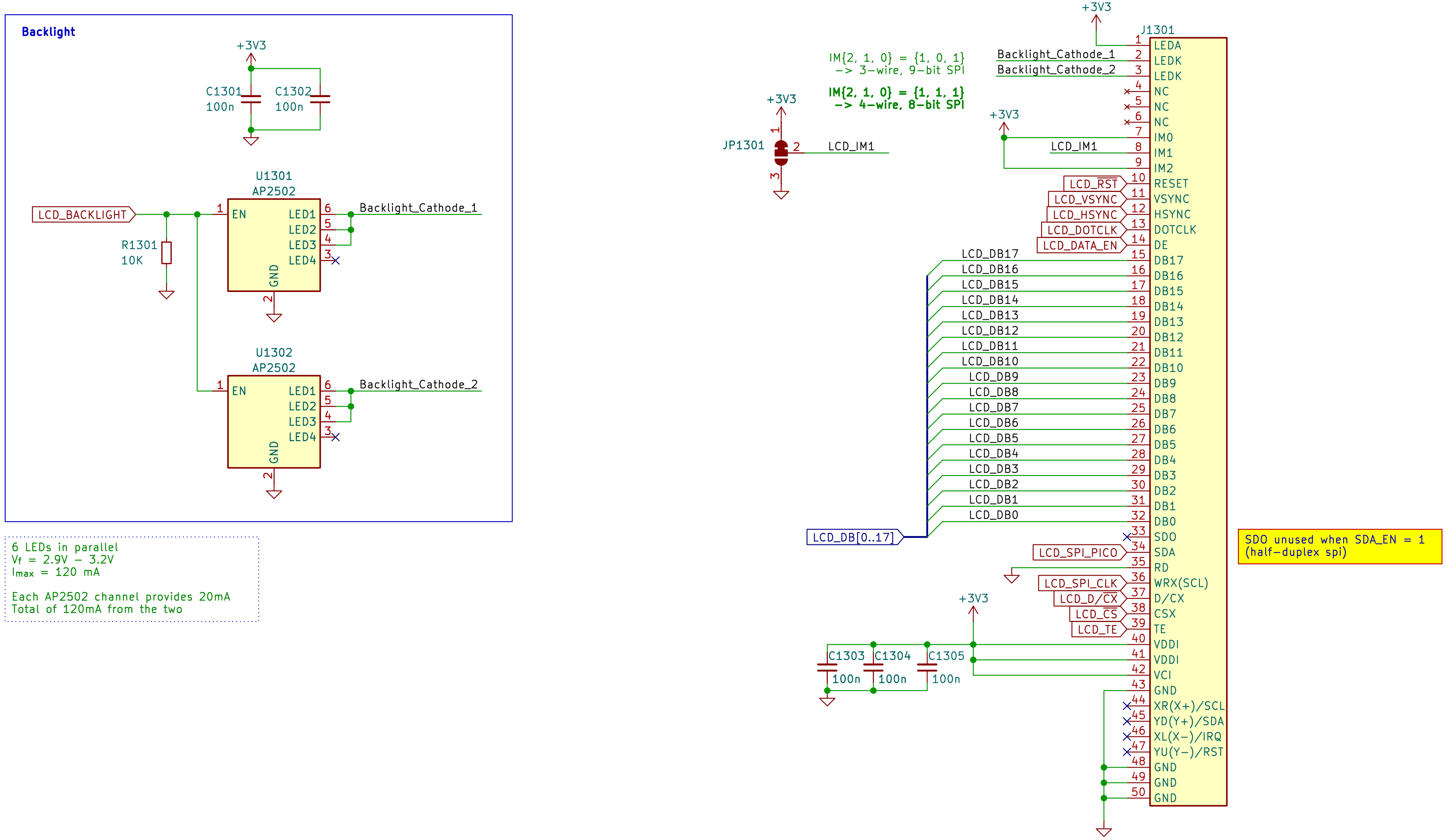


Sheet: /FPGA IO Bank 35/
File: fpga_io_35.kicad_sch

Title:

Size: A4 | Date: 2024-12-24 | **Rev: 2**

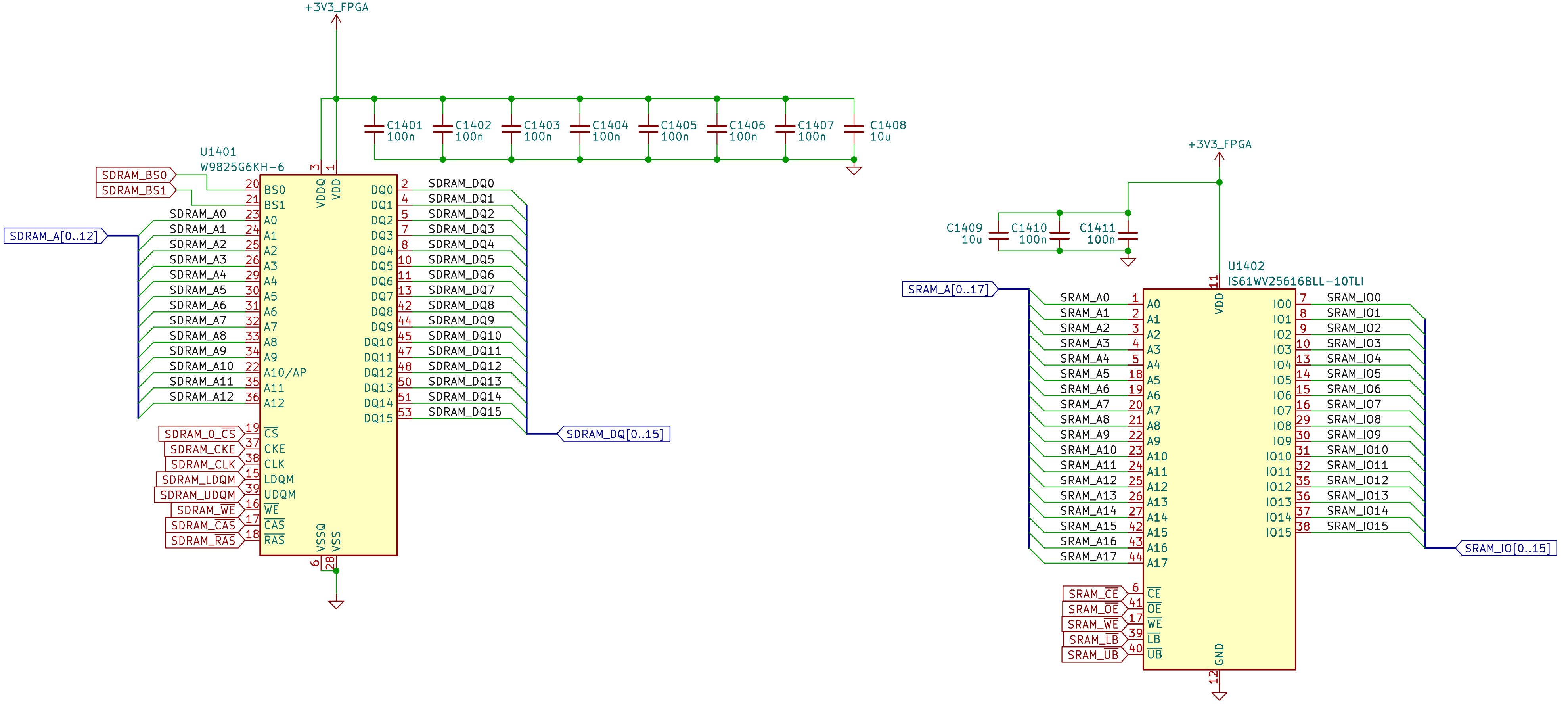
KiCad E.D.A. 8.0.7 | Id: 11/20



Sheet: /LCD/
File: lcd.kicad_sch

Title:

Size: A4	Date: 2024-12-24	**Rev: 2**
KiCad E.D.A. 8.0.7		Id: 12/20

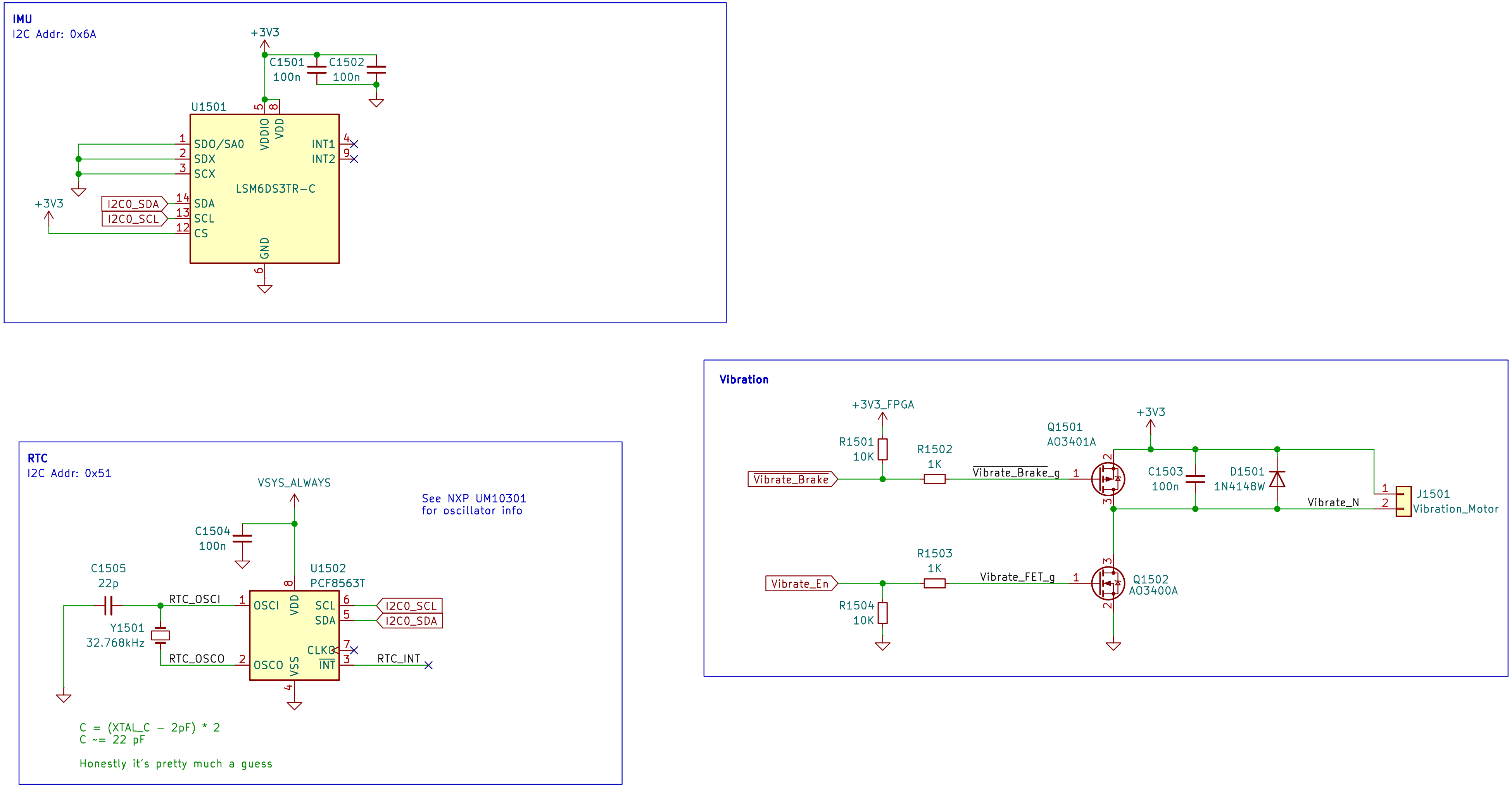


Sheet: /RAM/
File: ram.kicad_sch

Title:

Size: A4 | Date: 2024-12-24 | **Rev: 2**

KiCad E.D.A. 8.0.7 | Id: 13/20

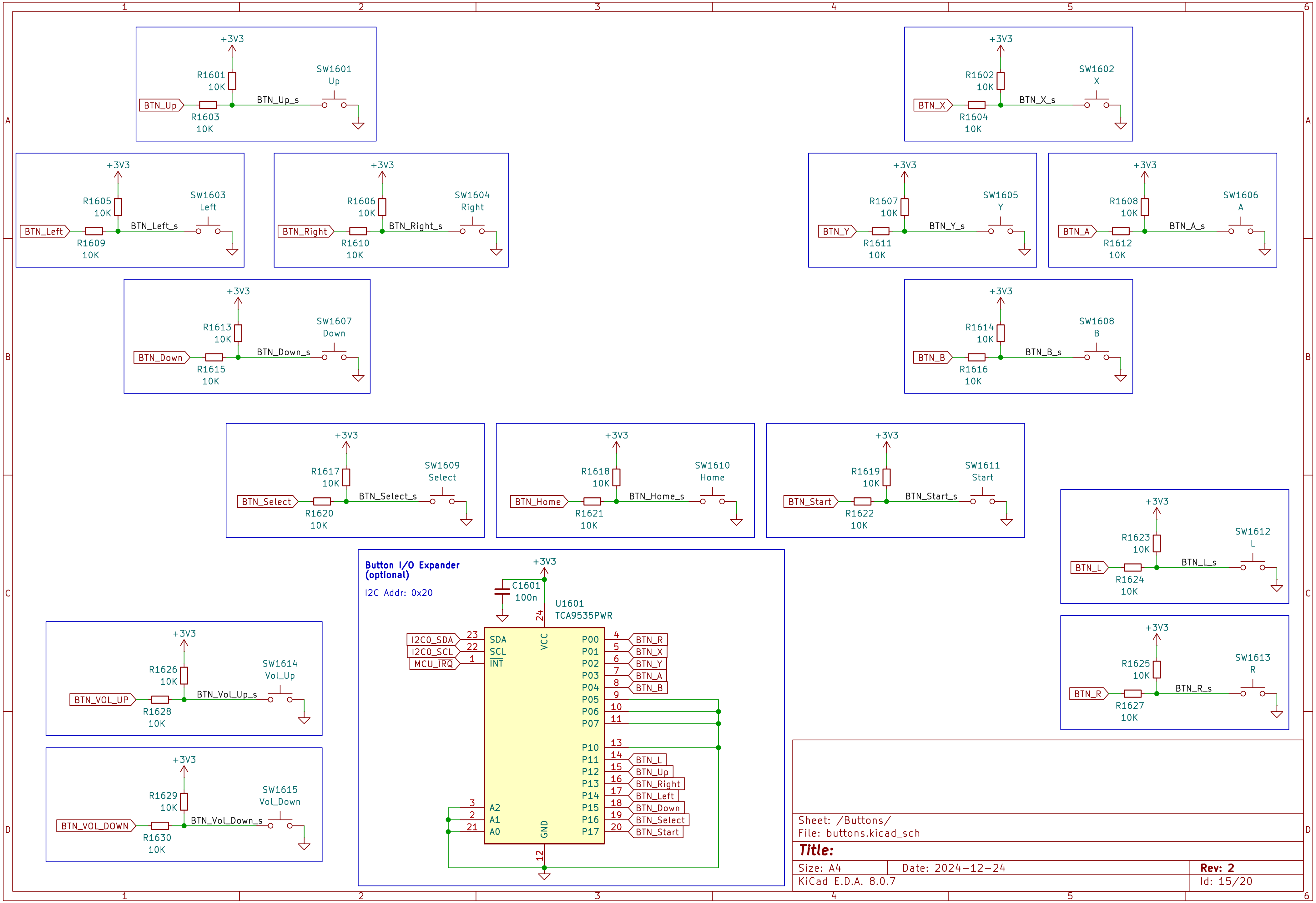


Sheet: /Peripherals/
File: peripherals.kicad_sch

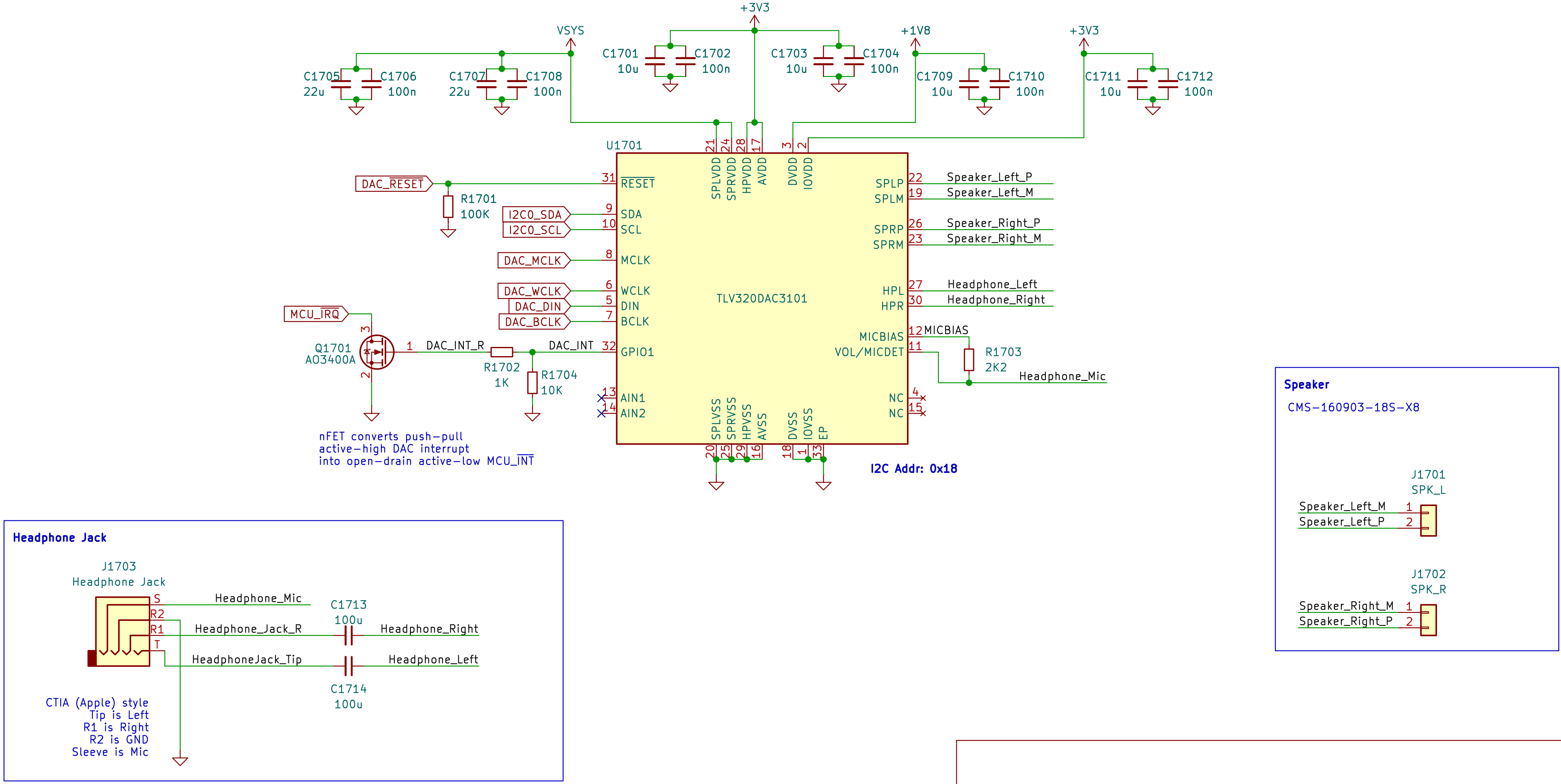
Title:

Size: A4 | Date: 2024-12-24 | **Rev: 2**

KiCad E.D.A. 8.0.7 | Id: 14/20



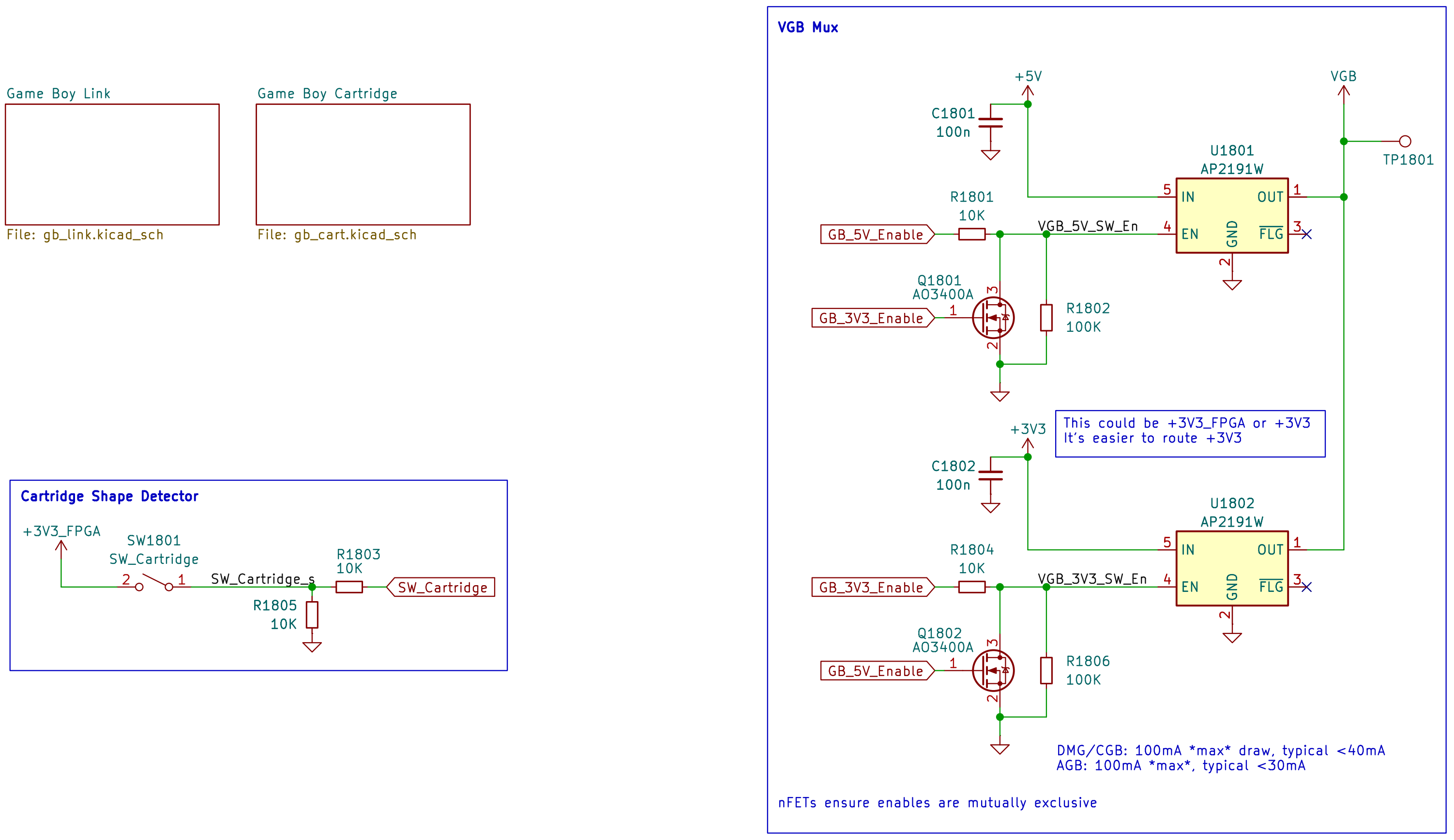
+3V3
R1601 10K
SW1601 Up
BTN_Up
BTN_Up_s
R1603 10K
+3V3
R1602 10K
SW1602 X
BTN_X
BTN_X_s
R1604 10K
+3V3
R1605 10K
SW1603 Left
BTN_Left
BTN_Left_s
R1609 10K
+3V3
R1606 10K
SW1604 Right
BTN_Right
BTN_Right_s
R1610 10K
+3V3
R1607 10K
SW1605 Y
BTN_Y
BTN_Y_s
R1611 10K
+3V3
R1608 10K
SW1606 A
BTN_A
BTN_A_s
R1612 10K
+3V3
R1613 10K
SW1607 Down
BTN_Down
BTN_Down_s
R1615 10K
+3V3
R1614 10K
SW1608 B
BTN_B
BTN_B_s
R1616 10K
+3V3
R1617 10K
SW1609 Select
BTN_Select
BTN_Select_s
R1620 10K
+3V3
R1618 10K
SW1610 Home
BTN_Home
BTN_Home_s
R1621 10K
+3V3
R1619 10K
SW1611 Start
BTN_Start
BTN_Start_s
R1622 10K
+3V3
R1623 10K
SW1612 L
BTN_L
BTN_L_s
R1624 10K
+3V3
R1625 10K
SW1613 R
BTN_R
BTN_R_s
R1627 10K
+3V3
R1626 10K
SW1614 Vol_Up
BTN_VOL_UP
BTN_Vol_Up_s
R1628 10K
+3V3
R1629 10K
SW1615 Vol_Down
BTN_VOL_DOWN
BTN_Vol_Down_s
R1630 10K
Button I/O Expander
(optional)
I2C Addr: 0x20
+3V3
C1601 100n
U1601 TCA9535PWR
24 VCC
I2C0_SDA 23 SDA
I2C0_SCL 22 SCL
MCU_IRQ 1 INT
3 A2
2 A1
21 A0
12 GND
P00 4 BTN_R
P01 5 BTN_X
P02 6 BTN_Y
P03 7 BTN_A
P04 8 BTN_B
P05 9
P06 10
P07 11
P10 13
P11 14 BTN_L
P12 15 BTN_Up
P13 16 BTN_Right
P14 17 BTN_Left
P15 18 BTN_Down
P16 19 BTN_Select
P17 20 BTN_Start
Sheet: /Buttons/
File: buttons.kicad_sch
Title:
Size: A4
Date: 2024-12-24
Rev: 2
KiCad E.D.A. 8.0.7
Id: 15/20



Sheet: /Audio/
File: audio.kicad_sch

Title:

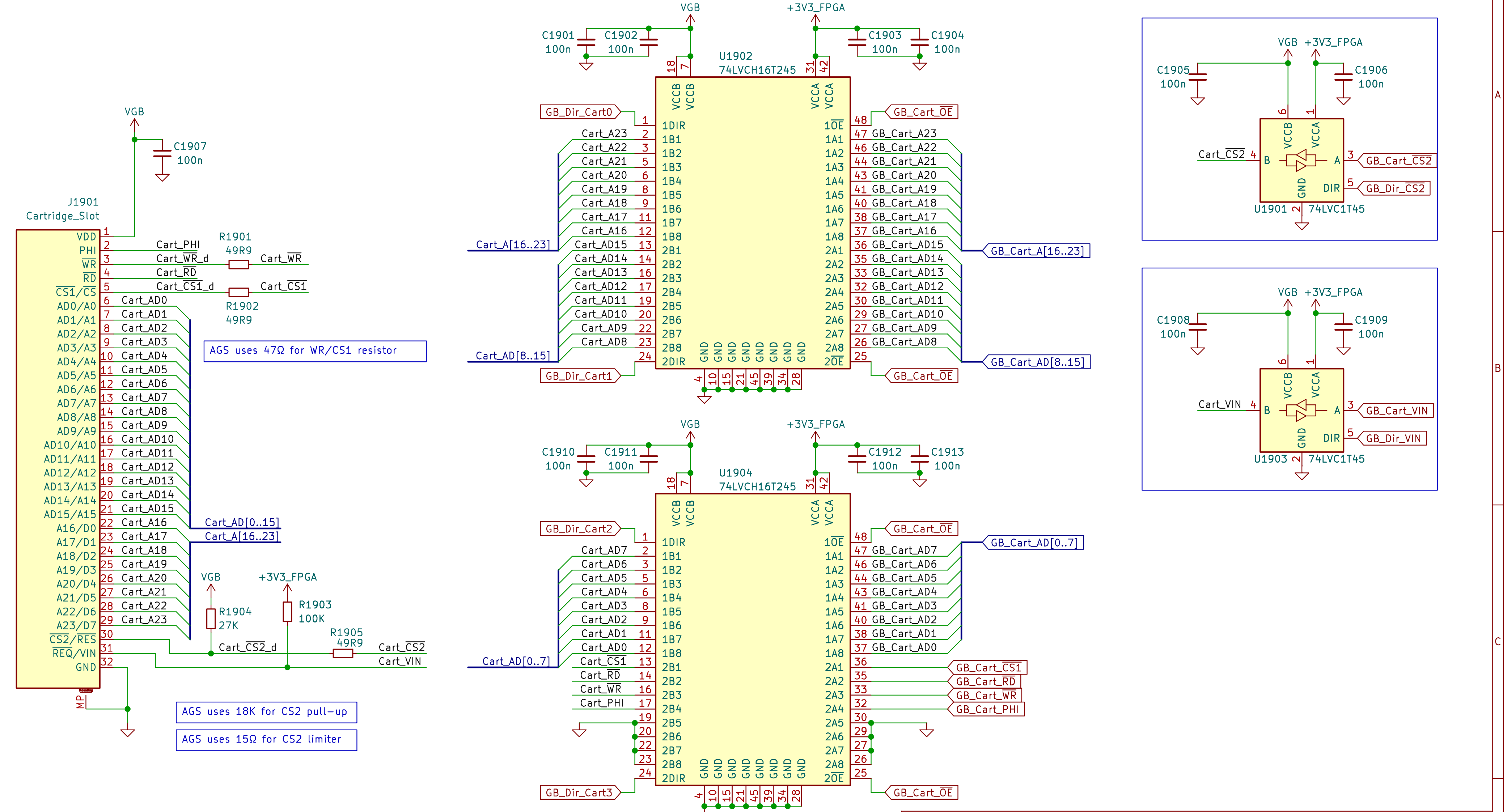
Size: A4	Date: 2024-12-24	Rev: 2
KiCad E.D.A. 8.0.7		Id: 16/20



Sheet: /Game Boy Interfaces/
File: gb_interfaces.kicad_sch

Title:

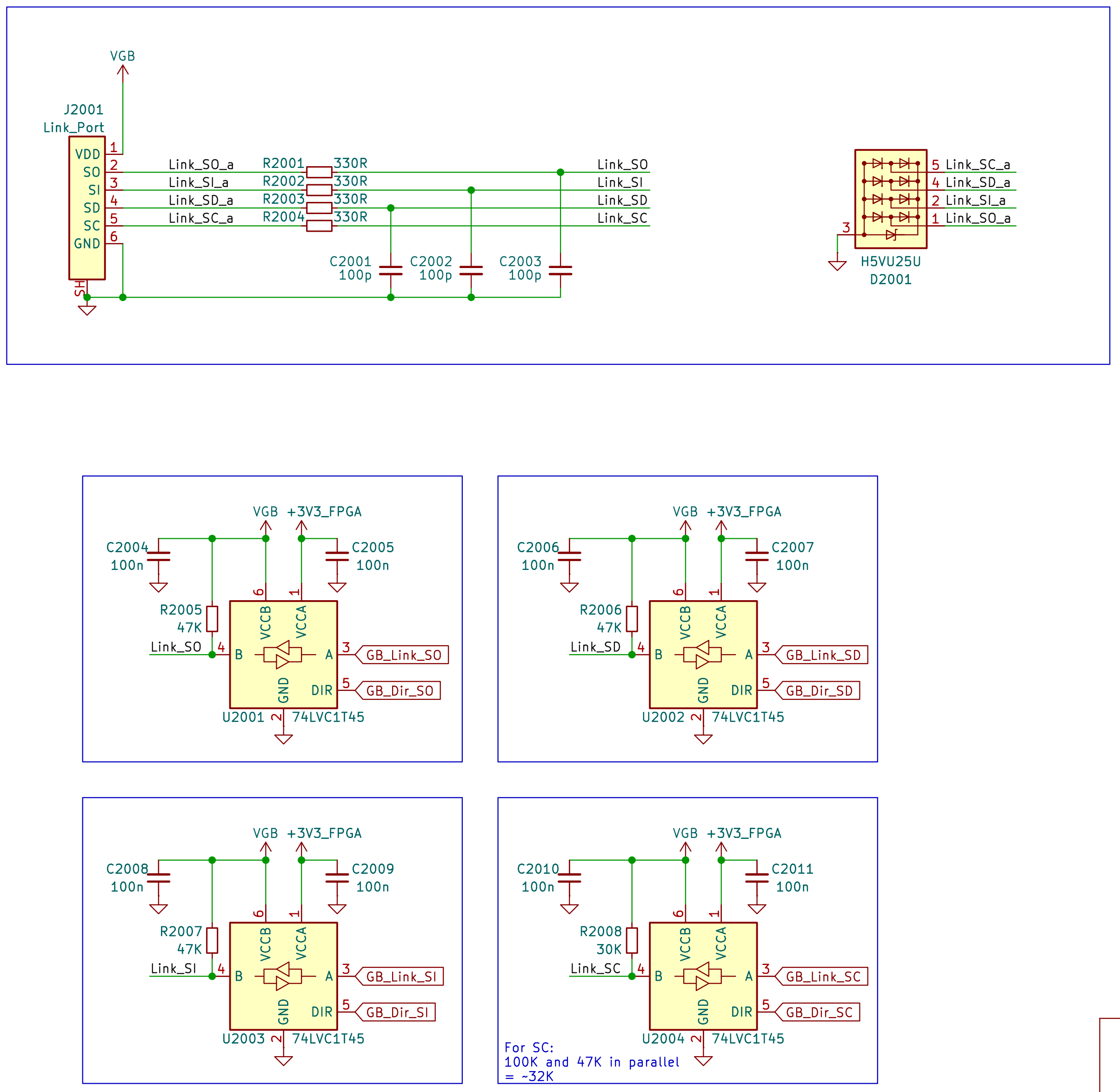
Size: A4	Date: 2024-12-24	**Rev: 2**
KiCad E.D.A. 8.0.7		Id: 18/20



Sheet: /Game Boy Interfaces/Game Boy Cartridge/
File: gb_cart.kicad_sch

Title:

Size: A4	Date: 2024-12-24	**Rev: 2**
KiCad E.D.A. 8.0.7		Id: 19/20



Sheet: /Game Boy Interfaces/Game Boy Link/
File: gb_link.kicad_sch

Title:

Size: A4	Date: 2024-12-24	Rev: 2
KiCad E.D.A. 8.0.7		Id: 20/20